UNIVERSITY OF CENTRAL PUNJAB

Faculty of Information Technology

Cyber Security

Week 1 — Introduction to Cybersecurity

Overview • Importance • CIA Triad
Basic Terminology • History of Cyber Threats

Instructor: Hafiz Usama Ishtiaq

BSCS • 3 Credit Hours • Fall 2026

WEEK 1 ROADMAP

What we will cover this week

1

Overview of Cybersecurity

What cybersecurity is, in simple words, and why it matters today

2

The CIA Triad

Confidentiality, Integrity, Availability — the three goals of security

3

Basic Terminology

Threat, vulnerability, risk, exploit, attack, asset — explained with examples

4

History of Cyber Threats

How attacks grew from harmless experiments to today's global threats

Grading (Tentative)

ITEM	WEIGHTAGE
Quizzes	15 %
Assignments	15 %
Class Participation	10%
Mid Term	20 %
Final Term	40 %
Total	**100**

Logistics

- Quiz may be un-announced.
- Read the book
- Plagiarism – Write your own Solution
- Deadline of the assignments will not be extended
- No Late submissions will be accepted.

Other Announcements

- Email address:
  - Usama.Ishtiaq@ucp.edu.pk
- Office:
  - C Block, Office 011.

OVERVIEW

What is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, programs and data from digital attacks, unauthorized access, damage or theft.

In simple words: it is everything we do to keep our digital life safe — just like we lock our house, we also need to "lock" our data and devices.

Protects

People, information and devices across websites, networks, software and mobile phones

Ensures

That your data stays private, correct, and available whenever you need it

Defends

Against dangers like viruses, hacking, phishing emails and data theft

IMPORTANCE

Why Cybersecurity Matters

Everything is connected

Banking, hospitals, universities and government all run on digital systems now.

Data is valuable

Your passwords, photos, bank details and ID are exactly what attackers want.

Attacks are rising

Cybercrime happens every single day and costs the world billions each year.

Trust & reputation

One data breach can make customers lose trust in a company forever.

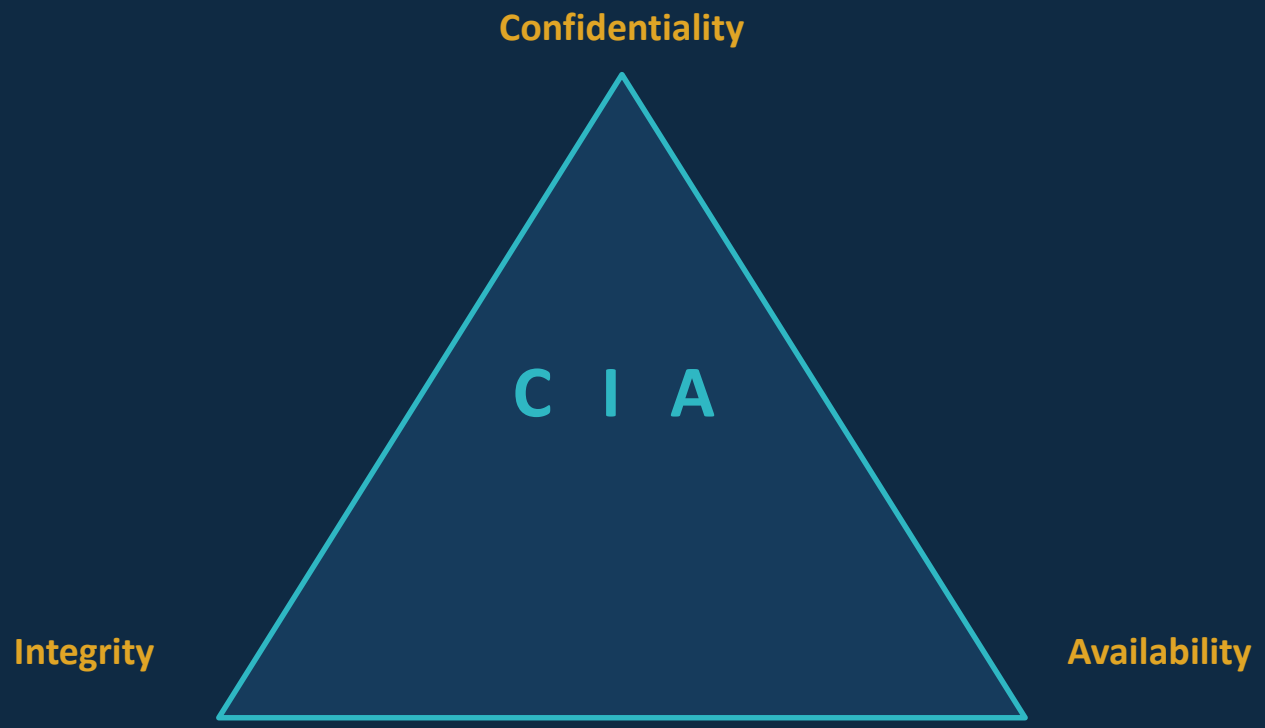
Legal responsibility

Laws require organizations to protect user data — and report it if leaked.

Safety of society

Power grids, hospitals and transport must stay secure to keep people safe.

The CIA Triad — 3 Goals of Security



These three goals work together. If even one fails, the system is not truly secure — so security means protecting all three at the same time.

CIA TRIAD

Confidentiality, Integrity, Availability

Confidentiality

Keeping data private

Only the right people can see the information. It is kept secret from everyone else.

EXAMPLE

Your exam result is shown only to you — not to the whole class.

Protected by: Encryption, passwords, MFA

Integrity

Keeping data correct

Information is accurate and no one has secretly changed it.

EXAMPLE

A bank makes sure nobody can change your account balance from Rs.5,000 to Rs.50,000.

Protected by: Hashing, checksums, backups

Availability

Keeping data usable

Systems and data are up and working whenever you need them.

EXAMPLE

An exam portal must stay online on results day so students can log in.

Protected by: Backups, redundancy, DDoS defense

BASIC TERMINOLOGY

Key Terms Every Student Must Know (1 of 2)

Asset	**Anything valuable worth protecting.** **Example:** Your laptop, phone, data, or a company's website and customer records.
Threat	**Anything that can cause harm.** **Example:** A hacker, a virus, a careless employee, or even a fire or power failure.
Vulnerability	**A weakness that can be misused.** **Example:** A weak password, outdated software, or an unlocked door to a server room.
Risk	**Chance of harm actually happening.** **Example:** High risk = a weak password (vulnerability) + an active hacker (threat).

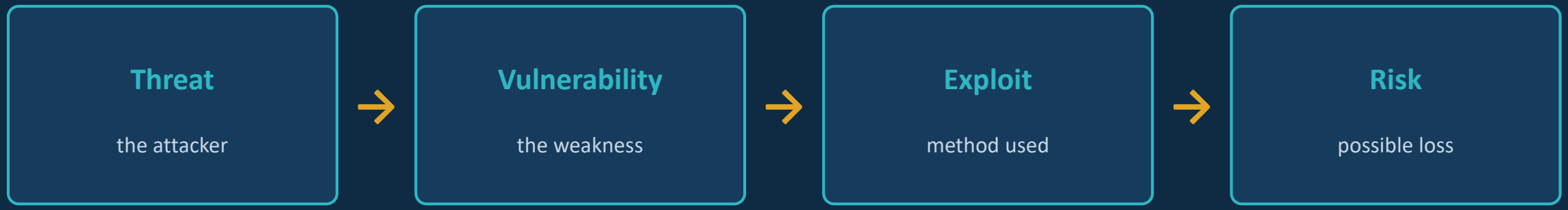
Key Terms Every Student Must Know (2 of 2)

Exploit	**A trick or code that uses a weakness.** **Example:** A program that breaks into software through a known bug.
Attack	**A deliberate attempt to cause harm.** **Example:** Sending a fake email to steal a user's login password.
Attack Vector	**The path used to get in.** **Example:** Email attachments, USB drives, fake websites, or public Wi-Fi.
Payload	**The harmful part that does the damage.** **Example:** The code that locks your files and demands ransom money.

PUTTING IT TOGETHER

How the Terms Connect

Read the chain left to right — this is how a real attack unfolds:



Real-life example

A hacker (threat) finds a website with an old, unpatched login page (vulnerability). **They use SQL injection** (exploit) to break in. The chance that customer data gets stolen is the **risk.**

How Cyber Threats Evolved

1970s–80s	**The first experiments** Creeper (1971) & Morris Worm (1988) — early self-copying programs, made out of curiosity, not for money.
1990s	**Viruses go mainstream** As home PCs and email spread, viruses spread with them — often through infected email attachments.
2000s	**Worms & organized crime** Fast worms like ILOVEYOU & Conficker spread worldwide; criminals started hacking to make money.
2010s	**Nation-states & ransomware** Governments joined in (Stuxnet); huge company data breaches; ransomware like WannaCry locked files for money.
2020s	**AI-era & supply chain** Ransomware-for-hire, attacks through trusted software vendors, and AI used to create smarter attacks.

Lessons From the Past

Attacks keep evolving

New tricks appear all the time, so defenders must keep learning. Yesterday's protection is not enough today.

The motive changed

Early hackers wanted fame or fun. Today it is mostly about money, spying, or political power.

The impact grew

It started with one computer. Now a single attack can hit millions of users and whole countries.

People are the weak point

Most modern attacks trick a person (e.g. a fake email) rather than the machine — so awareness is our best defense.

RECAP

Week 1 Summary

- Cybersecurity means protecting our computers, networks and data — like locking the doors of our digital life.
- It matters because everything is connected, our data is valuable, and attacks happen every day.
- The CIA Triad — Confidentiality, Integrity, Availability — is the three-part goal of all security.
- Core terms: asset (what we protect), threat (the danger), vulnerability (the weakness), risk (chance of harm), plus exploit, attack, attack vector and payload.
- Threats have grown from harmless 1980s experiments into organized, global, money-driven and AI-era attacks.

Identify the Security Terms

Scenario:

A university stores student records in an online database. A hacker discovers that the university server is running an **old version of software with a known security weakness**. The hacker sends a specially crafted request through the website and gains unauthorized access to the database. The hacker then copies students' personal information and changes some students' marks.

Questions for Students

Identify the following from the scenario:

1. **Asset:** What is being protected?
2. **Threat:** What is the danger?
3. **Vulnerability:** What is the weakness?
4. **Exploit:** What technique/action is used to take advantage of the weakness?
5. **Attack:** What has the hacker actually done?
6. **Attack Vector:** Through which route did the hacker enter?
7. **Payload:** What harmful action/data did the attack deliver?
8. **Risk:** What could happen to the university and students?
9. **CIA Triad:** Which of **Confidentiality, Integrity, Availability** are affected?

Term	Answer
Asset	Student records/database
Threat	Hacker
Vulnerability	Old software with a known weakness
Exploit	Specially crafted request used to exploit the weakness
Attack	Unauthorized access to the database
Attack Vector	University website/server
Payload	Stolen personal information and changed marks
Risk	Data theft, privacy loss, and incorrect academic records
CIA	**Confidentiality** (data stolen) + **Integrity** (marks changed)

CHECK YOUR UNDERSTANDING

Review Questions

1. Define cybersecurity in your own words. Why is it important today?
2. Explain the CIA Triad. Give one real example for each of the three parts.
3. What is the difference between a threat, a vulnerability, and a risk?
4. Define exploit, attack vector and payload, each with a simple example.
5. Describe how cyber threats have changed from the 1980s to today.

WEEK 2 PREVIEW

Coming Next: Computer System Threats

- Introduction to computer system threats
- Types of threats: internal vs external
- Attack vectors and common cyber threats